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Meta-analysis

Enhanced protein intake on maintaining muscle mass, strength, and physical function in adults with overweight/obesity: A systematic review and meta-analysis

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SUMMARY

Background & aims: Weight loss in individuals with obesity and overweight leads to metabolic and health benefits but also poses the risk of muscle mass reduction. This systematic review and meta-analysis of randomized controlled trials aims to determine the initial protein amount necessary for achieving weight loss while maintaining muscle mass, strength, and physical function in adults with overweight and obesity.

Methods: Relevant literature databases, including Medical Literature Analysis and Retrieval System Online (Medline), Excerpta Medica (Embase), the Cumulative Index to Nursing and Allied Health Literature (CINHAL), and Web of Science, were electronically searched up to 15 March 2023. We examined the effect of additional protein intake on muscle mass, strength, and physical function in adults with overweight or obesity targeting weight loss. The risk of bias was assessed using the Cochrane RoB 2.0 tool. Results were synthesized using standardized mean differences (SMD) and 95% confidence intervals (CI) via a random-effects model.

Results: Forty-seven studies ($n = 3218$) were included. In the muscle mass analysis, twenty-eight trials with 1989 participants were encompassed. Results indicated that increased protein intake significantly prevents muscle mass decline in adults with overweight or obesity aiming for weight loss (SMD 0.75; 95% CI 0.41 to 1.10; $p < 0.001$). Enhanced protein intake did not significantly prevent decreases in muscle strength and physical function. An intake exceeding 1.3 g/kg/day is anticipated to increase muscle mass, while an intake below 1.0 g/kg/day is associated with a higher risk of muscle mass decline. The risk of bias in studies regarding muscle mass ranged from low to high.

Conclusions: Adults with overweight or obesity and aim for weight loss can more effectively retain muscle mass through higher protein intake, as opposed to no protein intake enhancement.

1. Introduction

Obesity represents a significant concern in the field of global healthcare. It is associated with a marked increase in mortality [1–5] and healthcare costs [3,6] and poses a risk of many disorders, including infections [7], cancer [8], and metabolic and cardiovascular diseases [9,10]. Since 1975, worldwide obesity has nearly tripled [11], with overweight and obesity-related mortality in most countries exceeding that caused by underweight [11]. An additional major issue linked to obesity is the loss of skeletal muscle mass or quality, which may result from reduced physical activity, skeletal muscle inflammation, leptin resistance [12], or impaired signaling

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Abbreviations

BCAA	branched-chain amino acids
CI	confidence intervals
FFM	fat-free mass
LBM	lean body mass
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RCT	randomized controlled trial
RoB	risk of bias was assessed according to the Cochrane Collaboration risk–of–bias tool
RR	risk ratio
SD	standard deviation

in the skeletal muscle mammalian target of rapamycin (mTOR) pathway [13] – the primary regulatory pathway for skeletal muscle protein synthesis [14]. Therefore, addressing and preventing obesity and overweight has become an immediate and critical global health priority with extensive socioeconomic implications.

In individuals with obesity and overweight, weight loss leads to metabolic and health benefits but also poses the issue of muscle mass reduction. Treatment guidelines, including those proposed by major medical and scientific societies, recommend a moderate weight loss of 5%–10% to achieve improvements in metabolic function and health outcomes [15]. A moderate 5% weight loss improves multi-organ insulin sensitivity and β-cell function, and an additional weight loss of 11%–16% further increases insulin sensitivity in muscle in cases of obesity [16]. Furthermore, intentional weight loss in these individuals is associated with improvements in various cognitive domains [17]. However, a potential drawback of weight loss is the accompanying loss of skeletal muscle mass, potentially accelerating sarcopenia development. Weight loss, achieved through an energy-reduced diet, decreases both fat and fat-free (or lean body) mass [18–20]. Nonelderly males and females with overweight and obesity experience reductions in skeletal muscle mass of about 2–2.5 kg and 1–1.5 kg per 10 kg of weight loss, respectively, with voluntary calorie restriction in the absence of a structured exercise program [21]. Therefore, the optimal approach to treat or prevent obesity and overweight, and their associated negative health outcomes, involves reducing fat mass while simultaneously preserving muscle mass, strength, and physical function.

The minimum protein intake required to achieve weight loss while maintaining muscle mass, muscle strength, and physical function in adults with overweight or obesity remains unclear. Strategies to mitigate the loss of skeletal muscle mass and strength during weight loss may include adequate protein consumption. Diets rich in protein show promise for both preventing and managing obesity, due to protein's ability to induce a greater diet-induced thermogenic response compared to carbohydrates and fats [22]. Additionally, high protein intake can help preserve lean body mass during weight loss, potentially counteracting a reduction in resting energy expenditure [22]. The technical review and position statement of the American Society for Nutrition and the North American Association for the Study of Obesity recommend moderate dietary energy restriction with a protein intake of 1.0 g per kilogram of body mass per day during dieting to prevent or ameliorate medical complications associated with obesity [23]. Furthermore, higher total dietary protein intakes (1.2–1.6 g/kg/day), compared to normal protein intakes (0.8 g/kg/day), have been shown to preserve lean mass and improve body composition during weight loss in young, middle-aged, and older adults [24–29]. However, the efficacy of increasing protein intake during weight loss in limiting the weight-loss-induced loss of muscle mass remains uncertain, as evidenced by conflicting results in randomized controlled trials (RCTs) reported in the literature [30–34].

Clarifying the effect of enhanced protein intake on clinical outcomes is vital for understanding the importance of nutritional intervention in overweight or obesity, particularly for those aiming for weight loss under medical care. This systematic review focuses on determining the initial protein amount necessary for achieving weight loss while maintaining muscle mass, muscle strength, and physical function in adults with overweight or obesity.

2. Methods

The study design was a systematic review of randomized clinical trials. This study was conducted by the guidelines of the Cochrane Handbook for Systematic Reviews of Interventions [35] and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 [36]. The procedures for identification, screening, data extraction, and analysis were established and unanimously agreed upon by all researchers involved in the study. The intended methods were documented and registered with the International Prospective Register of Systematic Reviews under registration number CRD42023402243, prior to initiating the literature search, data extraction, and analysis.

2.1. Eligibility criteria

The inclusion criteria were structured around the 'PICOS' framework, which stands for population, intervention, comparator, outcome, and study design, and were delineated as follows.

1. P: Adult participants of sex, aged 18 years or older with overweight or obesity assessed by body mass index (BMI) of 25 or more, encompassing both non-patients and patients from any setting, without geographic location restrictions.
2. I: enhanced protein consumption, whether accompanied by energy restriction and exercise or not with the primary intervention being increased protein intake through dietary or oral nutritional supplement means at any dose or level. There were no restrictions regarding the additional protein dose or the duration of the intervention protocol. Interventions involving omega-3 fatty acids, beta-hydroxy-beta-methylbutyrate, creatine, phosphatidic acid, or anabolic steroids were excluded.
3. C: no intervention (control) or placebo.
4. O: a primary outcome that included muscle mass parameter (e.g., appendicular muscle mass (AMM), fat-free mass (FFM), etc.), with secondary outcomes muscle strength (e.g., grip strength, etc.) and physical function (e.g., gait speed, etc.).
5. S: RCTs or cluster RCTs were considered when published as a journal article. The literature search also considered conference papers (but not conference abstracts) and theses.

2.2. Search strategy

A literature search was conducted to identify randomized controlled trials (RCTs), cluster-RCTs, and crossover trials investigating the effect of additional protein intake on muscle mass, muscle strength, and physical function in overweight or obesity adults aiming for weight loss. This search focused on sources written in English and published in peer-reviewed journals up to February 2023. Relevant literature databases, including the Medical

Literature Analysis and Retrieval System Online (Medline), Excerpta Medica Database (Embase), the Cumulative Index to Nursing and Allied Health Literature (CINHAL), and Web of Science, were electronically searched (Supplementary Table 1). The literature search was performed by a medical librarian. Additionally, hand searching was performed. The search strategies were designed to assess studies evaluating the impact of enhanced protein intake, with comparisons to no protein enhancement, placebo interventions, or no intervention.

2.3. Data extraction and outcome measures

Three independent reviewers conducted the initial screening by reviewing the titles and abstracts of articles using the Rayyan application (Qatar Computing Research Institute, Doha, Qatar). Each article was categorized as either included or excluded. The exclusion criteria included systematic reviews, animal models, qualitative studies, case reports, conference abstracts, observational studies, and articles without abstracts. In instances of disagreement, a fourth independent reviewer made the final decision, following the PRISMA guidelines. For those marked as included, full-text reviews were undertaken. One independent reviewer extracted the background information (such as region and studies involving healthy or sarcopenic individuals), the intervention period (specified in weeks), types of supplemental proteins, whether energy intake was matched between groups, the use of protein as a quantitative supplement, and nutritional interventions involving only protein or amino acids. Additionally, we recorded indicators and measures of muscle mass, muscle strength, and physical performance. Subsequently, three reviewers verified the accuracy of the extracted information.

2.4. Risk of bias assessment

The risk of bias was assessed using the Cochrane Collaboration's Risk-of-Bias Tool (RoB 2), implemented via an Excel tool by reviewers [35,37]. For each outcome, a RoB 2 table was created, detailing descriptions and judgments (low risk of bias, some concerns, high risk of bias) across the following domains for each study: 1. bias arising from the randomization process; 2. bias due to deviations from intended interventions; 3. bias due to missing outcome data; 4. bias in measurement of the outcome; 5. bias in selection of the reported result. Quality assessments were independently conducted by two reviewers. In cases of disagreement, a third independent reviewer resolved the issue. A risk of bias graph was generated using the robvis tool [38].

2.5. Statistical analysis

When studies utilized identical intervention types, comparators, and outcome measures, their results were pooled using a random-effects meta-analysis. Standardized mean differences were calculated for continuous outcomes, along with 95% confidence intervals (CI) and two-sided P values. In cases of missing data, authors of the primary studies were contacted. Mean values and standard deviations (SD) between baseline and outcome measures were extracted. If SD was not reported, calculations were performed following the equations in the Cochrane Handbook [35]. Heterogeneity was assessed using the I^2 statistic, with an I^2 of 50% considered indicative of substantial heterogeneity. If I^2 exceeded 50%, a random-effects analysis was used to combine the data. Potential publication bias was evaluated by visually inspecting funnel plots [39]. All statistical analyses were conducted using the Review Manager package [37].

3. Results

3.1. Selected studies

In the initial literature search across various databases and registers, 983 articles were identified. After removing 492 duplicates, 491 articles remained eligible for primary screening. Screening of article titles and abstracts led to the exclusion of 409 articles. Additionally, 54 articles were identified from potentially relevant systematic reviews found in database searches. Consequently, of the 136 articles assessed (491 initially eligible minus 409 excluded plus 54 added), 47 articles [30,40–84] were ultimately included in the qualitative synthesis (Fig. 1).

3.2. Characteristics of included studies

The 47 analyzed articles involved a total of 3218 participants, with 56% being women. Participants' mean or median ages ranged from 18 to 65 years. The earliest included study was published in 2000, and the most recent in 2022. Tables 1–3 provide an overview of these studies, categorizing them by indicators of muscle mass, muscle strength, and physical function, along with their characteristics. For muscle mass, the types of supplemental enhanced proteins included branched-chain amino acids (BCAA) [53,70,74], whey [30,41,47,50,52,54,60,61,64,67,68,76,78], and casein [41,46,58,63,81]. Thirty-four studies [27,30,40–46,48–55,57,59,60,62–64,66,67,69,71–73,75,80,82–84] matched energy intake between groups; 26 studies [30,40,41,46–48,50,52–54,57–61,63–65,67,68,70,74,76,78,79,81] used protein as a quantitative supplement; and 37 studies [27,40–44,46,47,49–51,53–55,57,61–66,68–71,73–84] focused exclusively on nutritional intervention with protein or amino acids. Energy restriction for weight loss varied from 400 to 750 kcal/day, with a ratio of 20–30% per day. Supplementary Table 2 provides more details on each study. Sample sizes ranged from 17 [72] to 151 [60] participants. The participants' baseline BMI ranged from 27 to 40. Furthermore, after the intervention, the range of weight loss was 0.6–13.3 kg and 1.1–12%. In studies assessing protein ingestion, daily total protein intake in the intervention groups varied from 0.79 to 1.69 g/kg/day (44% of studies between 0.79 and 1.29 g/kg/day, 56% at or above 1.30 g/kg/day), and from 0.53 to 0.97 g/kg/day in placebo/control groups. Eight studies [30,44,46,48,59,65,67,84] combined resistance training with protein enhancement.

3.3. Outcomes

The assessment of muscle mass included indicators such as appendicular muscle mass [30], fat-free mass (FFM) [45,47,50,54,55,58,61,63,65,66,75,76,78,80–82,84], lean body mass (LBM) [40–42,46,57,60,74,79,83], and lean mass [43,44,48,49,51–53,59,62,64,67–73,76,77] (Table 1). Methods to measure muscle mass encompassed bioimpedance [44,47,50,51,54,55,61,74,78–80], dual-energy X-ray absorptiometry [27,30,40,41,43,48,49,52,53,57,59,60,62,64–71,75,77,82–84], magnetic resonance imaging [57,76], hydrodensitometry [72], skinfold measurements [46], total body water method [58,63,81], and whole-body air plethysmography (BodPod®) [42,45,73]. Muscle strength was assessed using indicators like bench press [65,84], chest press [46], hand grip strength [30,47,73,74,83], knee strength [56,68,76], lat pulldown [84], leg strength [41,44,46,48,59,67,76], shoulder press [46], and three exercises [71] (Table 2). Indicators for physical function included the 400 m walk [30,41,47], 6-min walk [42,48], maximal oxygen uptake [61], balance and functional capacity [65], chair stand [30,48], gait speed [30], sit-to-stand [30,56], short physical performance battery [41,42,47,67,73,74], stair-climbing,

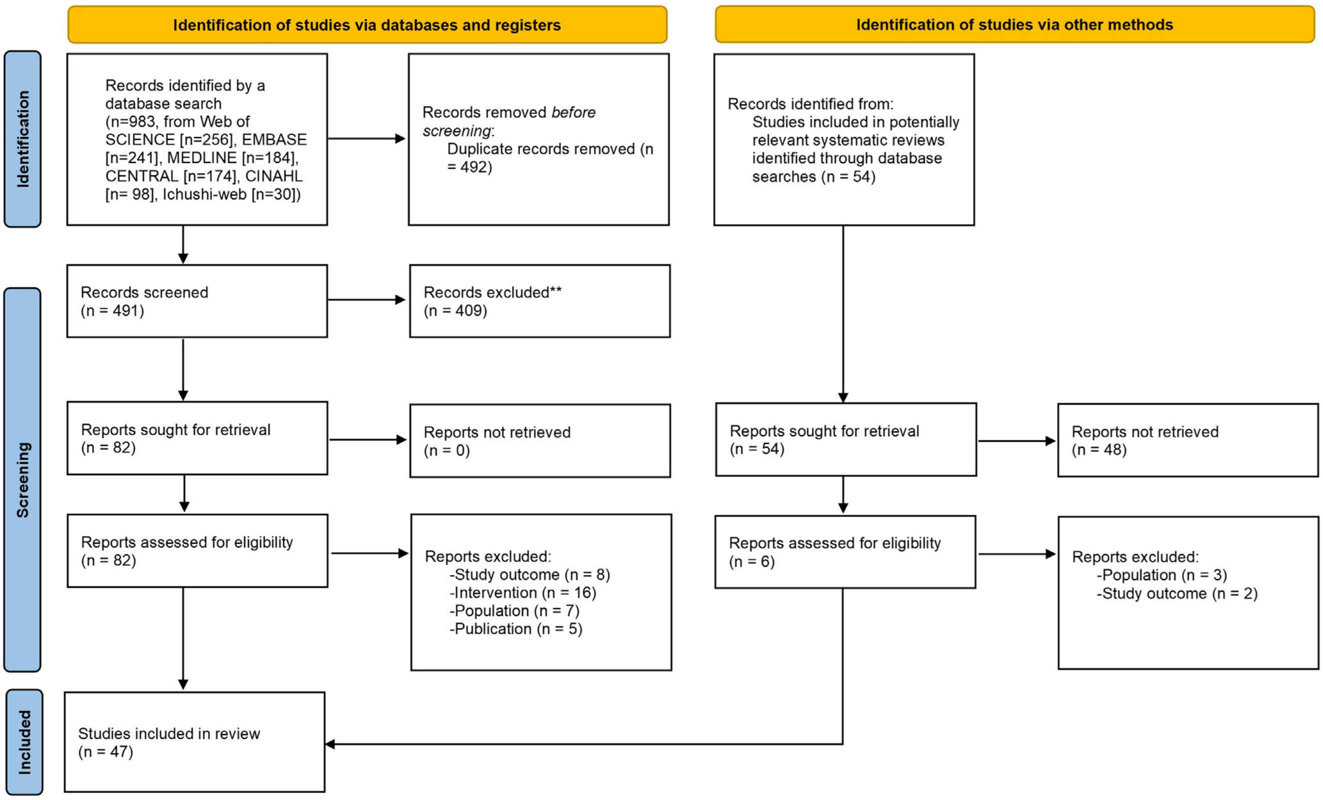

Fig. 1. Flowchart of Article Selection, based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Statement (n = number of studies).

Table 1
Number of studies on muscle mass indicators and their study design.

Muscle mass assessment indicator	Background						Intervention								
	Region				Healthy individuals study	Sarcopenia individuals study	Period (week)			Types of supplemental proteins			Energy intake matched between groups	Protein as a quantitative supplement	Nutritional intervention with protein or amino acids only
	Europe	North and Central America	South America	Asia and Oceania			<12	12 to 24	25<	BCAA	Whey	Casein			
Appendicular muscle mass	1	0	0	0	1	0	0	1	0	0	1	0	1	1	0
Fat free mass	8	4	2	3	13	0	7	9	1	0	6	3	10	10	15
Lean body mass	3	6	0	1	9	1	1	9	0	1	2	2	8	7	9
Lean mass	3	10	0	5	15	0	1	14	3	2	4	0	15	8	13

and up and go [48] (Table 3). All outcomes were measured as continuous data.

3.4. Quantitative synthesis

In the quantitative synthesis, findings from the included studies were analyzed and summarized. Intervention effects were presented using standardized mean differences (SMD) as the summary statistic, selected due to the varied methods used in these studies to measure outcomes such as appendicular muscle mass, FFM, bench press, handgrip strength, 400 m walk, and 6-min walk. Our meta-analysis of muscle mass encompassed twenty-eight trials with 1989 participants. It indicated that increased protein intake significantly prevents muscle mass decline in adults with overweight or obesity aiming for weight loss (SMD 0.75; 95% CI 0.41 to 1.10; $p < 0.001$; $I^2 = 92\%$; Fig. 2). Muscle strength was assessed in fourteen trials involving 854 participants. The meta-analysis showed that enhanced protein intake did not significantly prevent a decrease in muscle strength (SMD 0.01; 95% CI −0.13 to 0.14; $p = 0.92$; $I^2 = 0\%$; Fig. 2). Regarding physical function, fifteen trials with 1077 participants were included. The meta-analysis indicated that increased protein intake did not significantly prevent a decline in physical function (SMD 0.32; 95% CI −0.02 to 0.67; $p = 0.07$; $I^2 = 87\%$; Fig. 2).

3.5. Risk of bias assessment

The risk of bias assessment for randomized controlled trials (RCTs) focusing on muscle mass, muscle strength, and physical function, conducted using the ROB-2 tool, is depicted in

Table 2
Number of studies on muscle strength indicators and their study design.

Muscle strength assessment indicator	Background						Intervention								
	Region				Healthy individuals study	Sarcopenia individuals study	Period (week)			Types of supplemental proteins			Energy intake matched between groups	Protein as a quantitative supplement	Nutritional intervention with protein or amino acids only
	Europe	North and Central America	South America	Asia and Oceania			<12	12 to 24	25<	BCAA	Whey	Casein			
Bench press	0	1	0	1	0	0	0	2	0	0	0	0	1	1	2
Chest press	0	1	0	0	1	0	0	1	0	0	1	1	1	1	1
Hand grip strength	3	1	0	1	4	1	0	5	0	0	2	2	3	3	4
Knee strength	1	1	0	1	3	0	0	3	0	0	2	0	1	2	3
Lat pulldown	0	0	0	1	0	0	0	1	0	0	0	0	1	0	1
Leg strength	2	5	0	0	6	0	0	7	0	0	4	2	6	6	4
Shoulder press	0	1	0	0	1	0	0	1	0	0	1	1	1	1	1
Three exercises	0	0	0	1	1	0	0	1	0	0	0	0	1	0	1

Table 3
Number of studies on physical function indicators and their study design.

Physical function assessment indicator	Background						Intervention								
	Region				Healthy individuals study	Sarcopenia individuals study	Period (week)			Types of supplemental proteins			Energy intake matched between groups	Protein as a quantitative supplement	Nutritional intervention with protein or amino acids only
	Europe	North and Central America	South America	Asia and Oceania			<12	12 to 24	25<	BCAA	Whey	Casein			
400 m walk	3	0	0	0	3	0	0	3	0	1	3	1	3	3	2
6-min walk	0	2	0	0	2	0	0	2	0	0	0	0	2	1	1
Maximal oxygen uptake	1	0	0	0	1	0	1	0	0	0	1	0	0	1	1
Chair stand	1	1	0	0	2	0	0	2	0	1	1	0	2	2	0
Gait speed	1	0	0	0	1	0	0	1	0	1	1	0	1	1	0
Sit-to-stand	0	0	0	1	1	0	0	1	0	0	0	0	1	0	1
Short physical performance battery	3	2	0	0	4	1	0	5	0	1	3	1	4	4	5
Stairs-climbing	0	0	0	1	1	0	0	1	0	0	0	0	1	0	1
Up and go	0	1	0	0	1	0	0	1	0	0	1	0	1	1	0

Supplementary Fig. 1. Among the 28 studies included in the muscle mass category, two were assessed as having a low risk of bias (64,67), 20 exhibited some concerns regarding bias (27,30,41,43,46,47,52,53,57,61,62,70,75,76,78,79,82–84), and six were considered to have a high risk of bias (42,59,60,71–73). In the muscle strength category, out of 13 studies, two had a low risk of bias (67), nine had some concerns of bias (30,41,47,68,83,84), and two were deemed to have a high risk of bias (59,73). For the physical function category, of the 14 included studies, three were evaluated as low risk (67), eight as having some concerns (30,41,47,61), and three as high risk of bias (42,73).

3.6. Additional analysis

Figure 3 demonstrates the relationship between the quantity of protein intake and changes in muscle mass. An intake exceeding 1.3 g/kg/day is anticipated to lead to an increase in muscle mass (represented by line A), while an intake below 1.0 g/kg/day is linked to a higher risk of muscle mass decline (represented by line B). Furthermore, the asymmetry observed in funnel plots (Supplementary Fig. 2) suggests the presence of publication bias in the reporting of physical function outcomes.

4. Discussion

Enhanced protein intake significantly prevents muscle mass decline in adults with overweight or obesity who are aiming for weight loss. Moreover, a protein intake exceeding 1.3 g/kg/day is expected to result in increased muscle mass, while an intake below 1.0 g/kg/day is associated with a higher risk of muscle mass decline. Identifying an optimal initial protein dosage that facilitates weight loss while preserving muscle mass in this demographic represents a notable advancement in nutritional intervention strategies. In contrast, increased protein intake did not significantly prevent decreases in muscle strength and physical function. This study provides new evidence on the relationship between enhanced protein intake and muscle mass, strength, and physical function in adults with overweight or obesity.

It has long been recognized that increased protein intake during energy restriction helps prevent muscle mass loss [85]. Several systematic reviews and meta-analyses have explored the relationship between protein intake and muscle mass maintenance. For instance, Wycherley et al. [86] in 2012 reported that a high-protein diet, when prescribed at equivalent energy levels for over 4 weeks as compared to a standard protein diet during energy

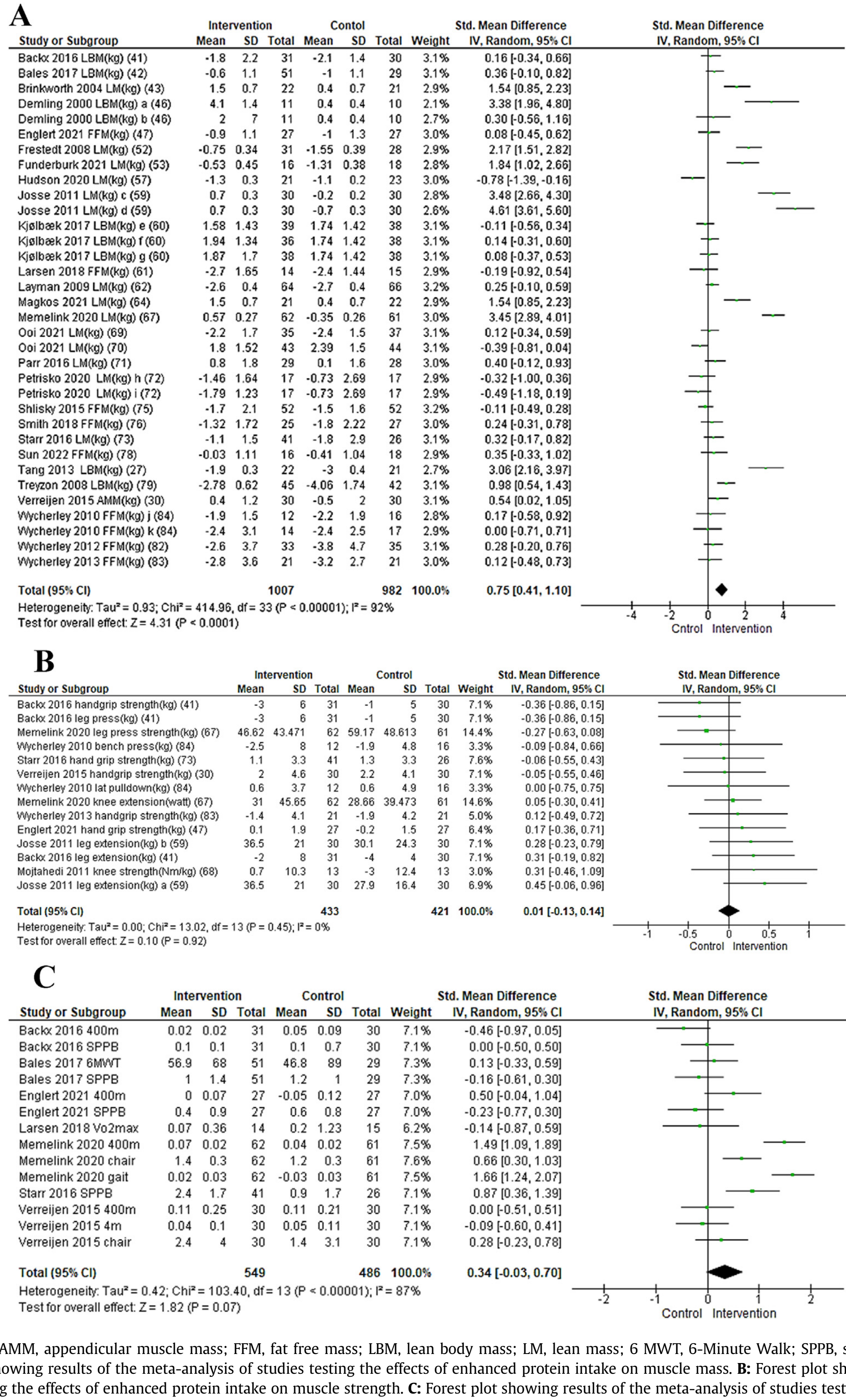

Fig. 2. Abbreviations – AMM, appendicular muscle mass; FFM, fat free mass; LBM, lean body mass; LM, lean mass; 6 MWT, 6-Minute Walk; SPPB, short physical performance battery. **A:** Forest plot showing results of the meta-analysis of studies testing the effects of enhanced protein intake on muscle mass. **B:** Forest plot showing results of the meta-analysis of studies testing the effects of enhanced protein intake on muscle strength. **C:** Forest plot showing results of the meta-analysis of studies testing the effects of enhanced protein intake on physical function. a CON vs Enhance (casein group); b CON vs Enhance (whey group); c CON (adequate protein, low dairy group) vs Enhance; d CON (adequate protein, medium dairy group) vs Enhance; e CON vs Enhance (calcium group); f CON vs Enhance (whey protein group); g CON vs Enhance (soy protein group); h CON (lower in fat and protein content) vs Enhance (very low in carbohydrate with animal sources of protein and fat group); i CON vs Enhance (very low in carbohydrate with mushroom- and plant-based group); j CON vs Enhance (high protein group); k CON vs Enhance (high protein and exercise); l CON (adequate protein, low dairy group) vs Enhance; m CON (adequate protein, medium dairy group) vs Enhance Figure 2: Effects of enhanced protein intake on changes in muscle mass, muscle strength and physical function. A: Muscle mass
B: Muscle strengthC: Physical functio.

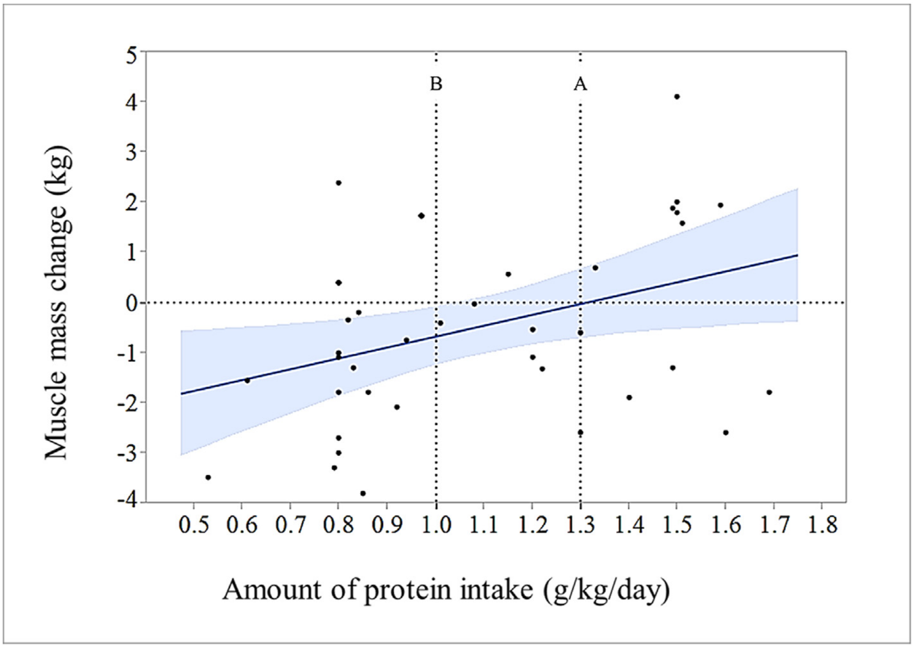

Fig. 3. Protein intake >1.3 g/kg/day is expected to increase muscle mass (line A), while <1.0 g/kg/day is at high risk of a decrease in muscle mass (line B) Fig. 3: Relationship between amount of protein intake and muscle mass change.

restriction, offered modest benefits in reducing FFM loss (FFM: 0.43 kg, 95% CI: 0.09 to 0.78). Similarly, Kim et al. [87] in 2016 demonstrated that protein intakes of ≥1.0 g/kg/day help preserve lean mass during successful weight loss in older adults with overweight or obesity. Additionally, a 2021 study by Tagawa et al. [88] found that total protein intake increases LBM in a dose-dependent manner across a wide range of doses. Our study's results align with these findings. However, it's important to note that previous systematic reviews and meta-analyses were not exclusively focused on overweight or obesity subjects. The distinct feature of our study is its specific focus on participants with overweight or obesity. In contrast, the 2020 systematic review and meta-analysis by Yin et al. [89] on sarcopenic obesity indicated that low-caloric high-protein diets did not show superiority over low-caloric low-protein diets in increasing FFM. Although our study included only one RCT targeting sarcopenic obesity, this RCT was not part of our meta-analysis. The variation in participant profiles between our study and that of Yin et al. [89] might account for the differences in findings. Our study specifically addresses non-sarcopenic obesity.

The findings of this study suggest that maintaining protein intake at or above 1.3 g/kg/day could potentially lead to an increase in muscle mass, while intake below 1.0 g/kg/day may increase the risk of muscle loss. A systematic review and meta-analysis of ICU patients [90] found a positive association between high protein provision and changes in muscle mass in critically ill patients, especially when comparing intakes of ≥1.2 g/kg/day versus <1.2 g/kg/day. Additionally, a systematic review and meta-analysis focusing on men and women aged 50 years and older [87] supported the role of consuming ≥1.0 g/kg/day of protein in preserving lean mass during successful weight-loss interventions. Furthermore, a systematic review and meta-analysis in non-obesity healthy adults [91] demonstrated significant impacts on lean body mass (LBM) in subjects ≥65 years old consuming 1.2–1.59 g of protein/kg/day and in younger subjects (<65 years old) consuming ≥1.6 g of protein/kg/day with resistance training. The results of our study, showing increased muscle mass with protein intakes greater than 1.3 g/kg/day and elevated risk of muscle loss with intakes less than 1.0 g/kg/day, align with these findings in non-obesity populations. Identifying an optimal initial protein amount that facilitates weight loss while preserving muscle mass in individuals with overweight or obesity is a significant advancement in nutritional intervention for this demographic.

The analysis showed a higher protein intake (above 1.3 g/kg/day) significantly prevents muscle mass decline. Maintaining muscle mass during weight loss can help mitigate the risk of sarcopenic obesity. The negative clinical consequences of sarcopenic obesity are of paramount importance. Sarcopenic obesity has been consistently demonstrated to be a strong and independent risk factor for frailty, comorbidities, and mortality in various highly prevalent disease conditions, as well as mortality [92], hence prevention and therapeutic intervention are important. Maintaining muscle mass during weight loss may help mitigate the risk of sarcopenic obesity, and improve overall physical function and mortality.

Enhanced protein intake was found to have no significant effect on maintaining muscle strength and physical function. Systematic reviews and meta-analyses in subjects with sarcopenic obesity have shown mixed results. For example, a 2022 study [93] reported that protein supplementation can effectively increase grip strength, while a 2019 study [94] found that low-calorie high-protein diets did not improve grip strength. Moreover, combining exercise with dietary supplements has been reported to increase grip strength and gait speed [95], but the effects of enhanced protein intake alone remain uncertain. In a recent systematic review and meta-analysis of non-obesity healthy adults [91], increased protein intake slightly improved bench press strength in subjects under 65 during resistance training, but its effects on handgrip strength and physical function test performance were minimal. These findings suggest that the impact of protein alone on maintaining or improving muscle strength and physical function is limited. Additionally, some studies indicate that while weight loss may reduce muscle mass, it does not significantly affect muscle strength [96]. Hence, unlike muscle mass, muscle strength may not substantially decrease with weight loss, and the beneficial effect of enhanced protein intake on muscle strength may not be as pronounced.

This study has several limitations. Firstly, the strength of our conclusions is tempered by the presence of some concerns or a high risk of bias related to study attrition in the analyzed studies. Secondly, most of the studies in this meta-analysis (44 out of 48) had intervention periods shorter than 25 weeks. Consequently, the long-term effects of these interventions remain uncertain, underscoring the need for additional research on longer-term outcomes. Thirdly, there was a suspected publication bias in the physical function results. Efforts to mitigate this included searching for unpublished references and contacting original authors, including those listed in trial registries. Finally, variability in study design, such as types of nutritional interventions, methods of analysis, and the selection of primary/secondary outcomes, limited the ability to compare results across studies. By strictly defining the population characteristics and ensuring consistency in interventions and outcomes measured across studies, researchers can minimize variability and reduce heterogeneity in systematic reviews. This involves using similar types and amounts of protein intake and identical outcome measurement tools across studies. However, in the field of clinical nutrition, intervention studies will inevitably feature a diversity of intervention types, observation periods, and outcomes, making it difficult to standardize studies rigorously in some areas. Moreover, overly restrictive criteria may limit the number of studies available for analysis, potentially affecting the review's statistical power and generalizability. Despite these limitations, this systematic review and meta-analysis significantly contribute to understanding the impact of enhanced protein intake on adults with overweight or obesity undergoing weight loss. Future high-quality, long-term RCTs that comprehensively address the risk of bias are crucial to advance our understanding in this area.

5. Conclusion

The findings from this systematic review and meta-analysis suggest that adults with overweight or obesity and aiming for weight loss can more effectively retain muscle mass through increased protein intake, as opposed to no enhancement in protein intake. However, enhanced protein intake did not significantly prevent a decrease in muscle strength and physical function. These insights reinforce the scientific basis for recommending a minimum protein intake of 1.3 g/kg/day for overweight and obesity adults to help preserve muscle mass as part of an effective weight-loss intervention.

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Author contributions

The authors' responsibilities were as follows—YK: is the guarantor and drafted the manuscript; JU, YS and KM: developed the search strategy and provided statistical expertise; and all authors: contributed to the development of the selection criteria, the risk-of-bias assessment strategy, and data extraction criteria; and provided feedback. All authors read and approved the final manuscript.

Declaration of competing interest

All authors declare that they have no conflict of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.clnesp.2024.06.030.

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